

Ready to harvest? Spine colour predicts gonad index and gonad colour rating of a commercially important sea urchin

Benjamin Mos*, Symon A. Dworjanyan

National Marine Science Centre, Southern Cross University, Coffs Harbour, New South Wales, Australia

ARTICLE INFO

Keywords:

Density
Water exchange rates
Tripneustes gratilla
Carotenoid
Harvesting strategy

ABSTRACT

An important problem limiting productivity and profitability of sea urchin aquaculture is the inability of producers to predetermine if sea urchins are ready for harvest. One way to overcome the 10–100% of lost production due to inopportune harvest times is to use external morphological characteristics to determine harvest readiness. To determine whether spine colour may be a useful indicator of harvest readiness, we tested whether there are relationships between spine colour and the size and colour of gonads of the commercially important sea urchin *Tripneustes gratilla*. Adult *T. gratilla* were grown in all combinations of three densities and three seawater exchange rates, simulating a range of environmental conditions occurring in culture systems. After six weeks, we measured the size and colour of gonads using standard protocols, quantified spine colour using colour rating and RGB (red, green, blue) intensity, and tested for relationships among these variables using linear models. The models identified significant positive relationships between spine colour predictors and gonad index and gonad colour rating. Our models indicated higher gonad indices were associated with brighter coloured spines, whilst gonads with the highest colour ratings were associated with stronger orange or red coloured spines. Patterns in spine colour identified by our models may reflect variation in colour producing pigments, carotenoids and naphthoquinones, in the epithelium. It is not clear why colour pigments in spines may depend on the colour and size of gonads, but it may be that healthy sea urchins that can grow large gonads also have sufficient energy to obtain and store colour pigments. Alternatively, colour pigments may provide health benefits that facilitate greater gonad size. Overall, our findings suggest that spine colour may be a suitable proxy for determining the harvest readiness of *T. gratilla* and possibly other sea urchins, but improvement in the reliability of our models and full automation of colour assessments are required before this novel technique can be implemented at commercial scales.

1. Introduction

Sea urchins are in high demand for their gonads or ‘uni’. Tightening supply caused by fisheries declines is seeing substantial interest in the development of post-harvest enhancement and closed-lifecycle aquaculture (Andrew et al., 2002; Brown and Eddy, 2015). To be economically viable, sea urchin aquaculture ventures must maximise gonad yield and quality to match market requirements (Unuma, 2002; Whitaker et al., 1997). Although there has been substantial research done to improve gonad yield by manipulating diet and culture methods, reliable production of high quality uni remains a challenge for industry (e.g. Brown and Eddy, 2015; Heflin et al., 2016; Shpigel et al., 2018).

One impediment to consistent production of high quality sea urchin gonads is that producers cannot assess the size or quality of gonads prior to harvest. Currently 10–100% of urchins within a cohort may be

harvested at an inopportune time, with uni obtained from these urchins being low grade or unsaleable due to poor colour, small size, or stage of gametogenesis (e.g. Azad et al., 2011; Grosjean et al., 1998; James, 2006; Shpigel et al., 2006; Woods et al., 2008). The only option available to estimate harvest readiness is to sacrifice a sample group. This imposes time and economic costs on producers, and may be a poor predictor of the quality of individuals in the remainder of the cohort if culture conditions across a farm are variable. A non-lethal and practical method for determining whether individuals are ready to harvest is required.

Many animals use colour to signal health, sexual maturity, and reproductive condition (Cott, 1940; Endler, 1980; Lozano, 1994). In general, sexually mature animals in peak condition display the brightest colours, whilst immature or stressed conspecifics tend to have dull colouration. The strong link between colour and health and/or sexual

* Corresponding author at: National Marine Science Centre, Southern Cross University, PO Box 4321, Coffs Harbour, NSW 2450, Australia.

E-mail address: benjamin.mos@scu.edu.au (B. Mos).

<https://doi.org/10.1016/j.aquaculture.2019.03.010>

Received 16 November 2018; Received in revised form 21 January 2019; Accepted 8 March 2019

Available online 09 March 2019

0044-8486/ © 2019 Elsevier B.V. All rights reserved.

maturity has enabled the development of protocols that use colour to determine optimal harvest or spawning times. For example, external colour of the Atlantic salmon, *Salmo salar* is linked to the quality of fillets after harvest (Aksnes et al., 1986; Folkestad et al., 2008). Changes in the colour of gonads seen through the transparent carapace can be used to predict spawning of shrimp, *Penaeus aztecus* and *Penaeus setiferus* (Brown and Patlan, 1974). Despite this, the efficacy of colour as an indicator of harvest readiness has not been examined for many marine invertebrates, including sea urchins.

The colour of an animal (hue, brightness, saturation, contrast, etc.) is determined by the type and amount of colour pigments, such as carotenoids and naphthoquinones, in tissues or body structures (Fox and Vevers, 1960). For sea urchins, carotenoids are primarily stored in the gonads, but are also found in external structures including the test and spines (Matsuno and Tsushima, 2001). Naphthoquinones are found in the test and spines, and are often the most abundant colour pigments in these structures (Anderson et al., 1969). As carotenoids are used for essential biological functions and are difficult to acquire and/or store (Olson and Owens, 1998), sea urchins stressed by adverse environmental conditions or inadequate dietary intake of carotenoids tend to have pale gonads (e.g. Azad et al., 2011; Mos et al., 2016; Pearce et al., 2003) due to low carotenoid content (Shpigel et al., 2006). Environmental stressors may also alter the deposition of colour pigments in spines because, in general, the types of carotenoids found in gonads and spines are consistent (Matsuno and Tsushima, 2001; Tsushima, 2007). The possibility that factors affecting carotenoid provisioning in gonads also have similar effects on spine colour is underlined by the fact that spine colours and gonad indices are simultaneously more similar among individuals living in the same habitat than for conspecifics living in different habitats (e.g. Addis et al., 2015; Lewis and Storey, 1984).

We propose there is a relationship between spine colour and the weight and colour of gonads of sea urchins that could be employed for non-lethal assessment of harvest readiness of individuals. As a first step in testing our hypothesis, this study examined whether there are relationships between spine colour and gonad colour or gonad index for the sea urchin *Tripneustes gratilla*. This species is cultured commercially in Australia, grown as a biocontrol in Hawaii (Westbrook et al., 2015), and is an aquaculture candidate in the Philippines, Israel, and South Africa (Cyrus et al., 2015; Juinio-Menez et al., 2008; Shpigel et al., 2018). *T. gratilla* has highly variable spine colour that varies in response to the same environmental factors that influence the size and colour of gonads (e.g. habitat, temperature, Mos et al., 2016; Mos et al., 2015; Toha et al., 2015), but it is not known if spine colour and gonad characteristics (size, colour, etc.) are interdependent.

A well-known phenomenon in production systems is that individuals often experience substantially different environmental conditions within the same culture system, resulting in substantial variation in feeding, growth, gonad production, or survival (e.g. Dong et al., 2010; Jobling and Baardvik, 1994; Qi et al., 2016). Inter-individual variation in exposure to environmental conditions is likely to be greatest for benthic invertebrates like sea urchins due to their slow movements and behaviours, such as the formation of high density feeding fronts (Dean et al., 1984; Lauzon-Guay and Scheibling, 2007) that also form in production systems. To simulate inter-individual variation in exposure to a range of environmental conditions that may occur within a commercial production system, *T. gratilla* were grown in all combinations of three densities and three seawater exchange rates. After six weeks, we tested for relationships between the colour of the urchin's spines and the weight and colour of their gonads using linear models.

2. Materials and methods

2.1. Experimental animals

Tripneustes gratilla were cultured at the National Marine Science

Centre, Southern Cross University, Coffs Harbour, Australia (30°12.5'S, 153°16.1'E) using established protocols (Mos et al., 2011), and used in the experiment when they began to produce gonads (~70 mm test diameter, ~7 months of age). *T. gratilla* were grown in all combinations of three densities (1, 3, and 5 individuals per replicate, equivalent to 9, 23, and 46 individuals.m⁻² of total surface area) and three seawater exchange rates (0.3, 1.0, and 3.0 exchanges.h⁻¹, equivalent to 20, 67, and 200 mL.min⁻¹) for six weeks at 26–27 °C. Each treatment had five replicates consisting of 5-L rearing containers (180 mm Ø base, 200 mm H) with over-flow holes that maintained the water volume at 4.0 L. Replicates were randomly assigned positions in an indoor water bath (100 mm deep) to maintain stable water temperatures. Replicates were maintained under a 12:12 photoperiod ('cool white' fluorescent). Sea urchins were fed the seaweed *Sargassum* sp. daily to excess. Faeces were removed mechanically by constant aeration provided by a 25 mm air-stone (0.71 L.min⁻¹) and containers were cleaned weekly.

There was no significant difference in the diameter (PERMANOVA, $F_{8,44} = 1.54$, $p = .18$; mean = 71.1 mm ± 0.4 SE), weight ($F_{8,44} = 0.87$, $p = .5547$; mean = 107.7 g ± 1.6 SE), and external colouration (e.g. test + spines Strength of Red (S_{RED}): $F_{8,36} = 2.00$, $p = .08$, mean = 0.397 ± 0.002 SE) of the urchins at the beginning of the experiment.

2.2. Data collection

After six weeks, all urchins were weighed, photographed to quantify spine colour, and then dissected to assess gonad index and gonad colour. The aboral surface of the urchins and their gonads were photographed (Panasonic Lumix DMC-FT10) within a light box alongside a paint colour series from white to red (see Mos et al., 2016 for details). White balance was set to *auto* and digital images were stored as JPEG files (resolution 4320 × 3240 pixels, three channels of 24 bit RGB colour information). Gonad colour was subjectively quantified by one individual using a rating scale (1–5) corresponding to the nearest colour in the paint colour series (1 representing white, 5 intense orange, Mos et al. (2016)). Gonad indices (GI) for all urchins were calculated using the formula:

$$GI\% = (W_G/W_U) \times 100$$

where W_G was the wet weight of the gonads (g) and W_U was the wet weight of the urchin (g).

Spine colour was assessed from photographs using a subjective method (spine colour rating) and an objective method, the RGB Measure routine in ImageJ (Rueden et al., 2017), with the outcomes of one method blinded to the outcomes of the other method. In ImageJ, individual spines (20 per urchin) were randomly selected using a random dot generator, selecting the closest spine to each dot beginning at the centre of the urchin and spiralling outwards. Spine colour was subjectively quantified by one individual using a rating scale (1–11) corresponding to the nearest colour in the paint colour series (1 representing white, 11 red, see Fig. 3a, Mos et al., 2016). Where spines were multiple colours (e.g. red and white), the rating value was assigned based on the colour that covered the greatest proportion of the spine.

Spine colour was objectively quantified using the RGB Measure protocol in ImageJ to obtain maximum ($_{MAX}$), minimum ($_{MIN}$), and mean ($_{MEAN}$) intensity for red, blue, and green points for each of the 20 spines measured from selected portions of the images (i.e. single spines only). Mean gray intensity was obtained from RGB data converted to grayscale using the formula 0.299 Red + 0.587 Green + 0.114 Blue. Mean RGB data were used to calculate strength of red (S_{RED}), strength of green (S_{GREEN}), and strength of blue (S_{BLUE}) for each spine following Gillespie et al. (1987) and Mos et al. (2016):

$$S_{RED} = \text{Red}_{MEAN} / (\text{Red}_{MEAN} + \text{Blue}_{MEAN} + \text{Green}_{MEAN})$$

$$S_{\text{BLUE}} = \text{Blue}_{\text{MEAN}} / (\text{Red}_{\text{MEAN}} + \text{Blue}_{\text{MEAN}} + \text{Green}_{\text{MEAN}})$$

$$S_{\text{GREEN}} = \text{Green}_{\text{MEAN}} / (\text{Red}_{\text{MEAN}} + \text{Blue}_{\text{MEAN}} + \text{Green}_{\text{MEAN}})$$

For each RGB variable, values for the 20 spines were averaged for each urchin, and the mean used in subsequent statistical analyses. The proportion of white spines out of 20 was also measured for each sea urchin (hereafter Proportion). A spine was considered to be white if the maximum intensity for all three RGB colours was > 235 . In the RGB model, values from 235 to the maximum 255 produce shades of white.

2.3. Statistical analysis

Multiple linear regressions were used to examine the relationships between spine colour variables (spine rating, S_{RED} , S_{GREEN} , S_{BLUE} , Red_{MEAN} , $\text{Green}_{\text{MEAN}}$, $\text{Blue}_{\text{MEAN}}$, Red_{MAX} , $\text{Green}_{\text{MAX}}$, Blue_{MAX} , Red_{MIN} , $\text{Green}_{\text{MIN}}$, Blue_{MIN} , Proportion, $\text{Gray}_{\text{MEAN}}$) and gonad rating or gonad index using mean data for each urchin (total number of urchins = 126). Marginal tests and draftsman's plots were used to analyse the relationship between a single spine colour variable and gonad rating or gonad index, and check for collinearity among predictors. Spine colour variables with p values $> .01$ were excluded from further analysis (Table 1). We did not include any environmental variables as predictors (e.g. density, seawater exchange rate) because this data would be unavailable for individual sea urchins from commercial culture systems, and we were interested in understanding whether our models would find relationships between spine colour and gonad size and colour in the absence of this information.

Linear regressions were conducted using the distance-based linear

Table 1

Distance-based linear model (DISTLM) marginal tests for gonad colour rating and gonad index of *Tripteneustes gratilla* and selected spine colour variables.

Dataset	Variable	SS (trace)	F	p	Prop.
Gonad Colour Rating	Spine Colour Rating	4.12	8.80	0.0045	0.07
	S_{RED}	0.12	0.25	0.6161	< 0.01
	S_{GREEN}	3.72	7.89	0.0047	0.06
	S_{BLUE}	7.54	17.12	0.0001	0.12
	Red_{MEAN}	12.98	32.73	0.0001	0.21
	$\text{Green}_{\text{MEAN}}$	11.16	27.13	0.0001	0.18
	$\text{Blue}_{\text{MEAN}}$	9.70	22.92	0.0001	0.16
	Red_{MAX}	11.67	28.67	0.0001	0.19
	$\text{Green}_{\text{MAX}}$	9.94	23.61	0.0001	0.16
	Blue_{MAX}	8.23	18.93	0.0002	0.13
	Red_{MIN}	8.97	20.92	0.0002	0.14
	$\text{Green}_{\text{MIN}}$	6.61	14.75	0.0004	0.11
	Blue_{MIN}	1.29	2.63	0.1024	0.02
	Proportion	0.12	0.25	0.6327	< 0.01
	$\text{Grey}_{\text{MEAN}}$	12.17	30.19	0.0001	0.20
Gonad Index	Spine Colour Rating	10.56	2.80	0.0939	0.02
	S_{RED}	42.70	12.16	0.0012	0.09
	S_{GREEN}	1.01	0.26	0.6169	< 0.01
	S_{BLUE}	95.73	31.03	0.0001	0.20
	Red_{MEAN}	66.73	20.11	0.0001	0.14
	$\text{Green}_{\text{MEAN}}$	53.20	15.52	0.0001	0.11
	$\text{Blue}_{\text{MEAN}}$	89.50	28.55	0.0001	0.19
	Red_{MAX}	70.25	21.35	0.0001	0.15
	$\text{Green}_{\text{MAX}}$	48.42	13.97	0.0004	0.10
	Blue_{MAX}	68.59	20.76	0.0001	0.14
	Red_{MIN}	43.55	12.42	0.0005	0.09
	$\text{Green}_{\text{MIN}}$	36.11	10.13	0.0019	0.08
	Blue_{MIN}	33.65	9.38	0.0021	0.07
	Proportion	0.92	0.24	0.6276	< 0.01
	$\text{Grey}_{\text{MEAN}}$	63.13	18.86	0.0001	0.13

Selection procedure: BEST, selection criterion: AICc. 'Prop.' is the proportion of variation in the dataset explained by the variable. Variables with $p > .01$ are bold. S_{BLUE} = strength of blue, S_{GREEN} = strength of green, S_{RED} = strength of red. Proportion = the proportion of spines that were designated as white (see Methods for details).

modelling (DISTLM) with distance based redundancy analysis (dbRDA) routine of Primer 6 (Primer-E, Plymouth) with PERMANOVA+ extension (v.6.1.18) software (Anderson, 2001; Anderson et al., 2008; McArdle and Anderson, 2001). The BEST model selection routine with AICc selection criterion (a modified version of Akaike's Information Criterion) and the BIC selection criterion (Schwarz's Bayesian Information Criterion), based on 9999 permutations, were used to select the five best fitting models. As analyses run using BIC selection criterion produced similar models to the AICc selection criterion, only the AICc models are presented.

Spearman's rank-order correlations and Pearson's correlations analyses were used to explore the relationships between the RGB variables identified by DISTLM models and gonad rating or gonad index, respectively. We also tested for a relationship between gonad index and gonad colour rating using a Spearman's rank-order correlation. Assumptions for correlation were checked using values for skewness and kurtosis, Shapiro-Wilk tests, and residual plots. All tests indicated the data were normally distributed and homoscedastic, except for gonad rating which was not normally distributed. There were no substantial differences in the outcomes of analyses where outliers were excluded or included, so the most conservative results where outliers were included are presented. Spearman's and Pearson's correlation analyses were conducted in IBM SPSS Statistics (v24.0.0).

3. Results

Spine colour was correlated with gonad colour rating and gonad index of *T. gratilla*. Models testing spine colour predictors accounted for 25–28% of the variation in gonad colour rating, and 33–34% of the variation in gonad index (Table 2). All AICc values in the top five models for gonad colour rating and gonad index were similar, indicating that models for each dataset had similar predictive power ($\text{AICc}_{\text{wt}} \sim 0.20$ for all models, Table 2). All the top five models contained the same RGB predictors, with other predictors contributing little to the models (Tables 1, 2). Gonad rating was correlated with the spine colour predictors Red_{MIN} , $\text{Green}_{\text{MIN}}$, and $\text{Green}_{\text{MEAN}}$ (Table 2). Gonad index was correlated with the spine colour predictors S_{BLUE} , Red_{MAX} , and $\text{Green}_{\text{MAX}}$ (Table 2).

When the relationships between spine colour predictors and gonad colour rating or gonad index were explored individually, analyses indicated all relationships were positive (Fig. 1). Spearman's rank-order correlations indicated there were significant positive relationships between gonad rating and Red_{MIN} ($r_s = 0.347$, $p < .001$, Fig. 1A), gonad rating and $\text{Green}_{\text{MIN}}$ ($r_s = 0.305$, $p < .001$, Fig. 1A), and gonad rating and $\text{Green}_{\text{MEAN}}$ ($r_s = 0.402$, $p < .001$, Fig. 1A). Increases in minimum and mean of red and green intensity likely reflect an increase in the intensity of colours resulting from the mix of these two parameters, particularly brown, yellow, orange, and red. This suggests higher gonad colour ratings were associated with stronger orange or red colours in spines (Fig. 2).

Pearson's correlations indicated there were significant positive relationships between gonad index and S_{BLUE} ($r = 0.447$, $p < .001$, Fig. 1B), gonad index and Red_{MAX} ($r = 0.383$, $p < .001$, Fig. 1B), and gonad index and $\text{Green}_{\text{MAX}}$ ($r = 0.318$, $p < .001$, Fig. 1B). Increased intensity in all RGB parameters is associated with brighter colours or white, suggesting higher gonad indices were associated with brighter coloured and/or white spines (Fig. 2).

Gonad colour rating had a significant positive correlation with gonad index (Spearman's rank-order correlation: $r_s = 0.542$, $p < .001$, Fig. 3). This suggests large gonads also had strong orange colouration that is indicative of high-grade uni.

4. Discussion

The development of non-lethal techniques for determining harvest readiness of sea urchins is vital given 10–100% of each cohort is

Table 2
Model selection estimating the relationship between spine colour and the gonad colour rating or gonad index of *Tripneustes gratilla*. In each case, ‘predictors’ refers to the combination of explanatory variables included in the model. $\Delta AICc$ is the difference in AICc between the model and the best fitting model. AICc-weight ($AICc_{wt}$) is the probability that this model represents the best-fitting model among those considered. ‘Sum of $AICc_{wt}$ ’ refers to the sum of $AICc_{wt}$ scores for the given model and all better-fitting models. For each dataset, the five best models are presented. S_{BLUE} = strength of blue, S_{GREEN} = strength of green, S_{RED} = strength of red.

Dataset	Predictors	AICc	$\Delta AICc$	$AICc_{wt}$	Sum of $AICc_{wt}$	R^2
Gonad Colour Rating	Red _{MIN} , Green _{MIN} , Green _{MEAN}	−117.78	0	0.21	0.21	0.25
	Red _{MIN} , Green _{MIN} , Green _{MEAN} , Red _{MEAN}	−117.15	0.63	0.20	0.41	0.26
	Red _{MIN} , Green _{MIN} , Green _{MEAN} , Gray _{MEAN} , Spine Rating	−117.08	0.70	0.20	0.61	0.28
	Red _{MIN} , Green _{MIN} , Green _{MEAN} , Gray _{MEAN}	−117.08	0.70	0.20	0.81	0.26
	Red _{MIN} , Green _{MIN} , Green _{MEAN} , Spine Rating, Red _{MEAN}	−116.63	1.15	0.19	1.00	0.27
Gonad Index	S_{BLUE} , Red _{MAX} , Green _{MAX} , S_{RED}	126.75	0	0.20	0.20	0.34
	S_{BLUE} , Red _{MAX} , Green _{MAX} , S_{GREEN}	126.86	0.11	0.20	0.40	0.34
	Red _{MAX} , Green _{MAX} , S_{RED} , S_{GREEN}	127.10	0.35	0.20	0.60	0.34
	S_{BLUE} , Red _{MAX} , Green _{MAX} , Green _{MEAN}	127.26	0.51	0.20	0.80	0.33
	S_{BLUE} , Red _{MAX} , Green _{MAX} , Gray _{MEAN}	127.29	0.54	0.20	1.00	0.33

harvested at inopportune times (e.g. Azad et al., 2011; Grosjean et al., 1998; James, 2006; Shpigel et al., 2006; Woods et al., 2008). This problem may be overcome by using external morphological characteristics as indicators of the harvest readiness of individuals. We found for the economically important *Tripneustes gratilla*, there were significant positive relationships between spine colour measured as RGB intensity and the colour and size of gonads. Our results indicate that spine colour

is a suitable proxy for determining the harvest readiness of *T. gratilla* and possibly other sea urchins, although further development is required before this novel technique can be trialled at commercial scales. We found *T. gratilla* with bold orange or red coloured spines tended to have gonads with intense orange colouration. There are strong relationships between the concentration of colour pigments in external and internal tissues across a wide variety of organisms, including birds

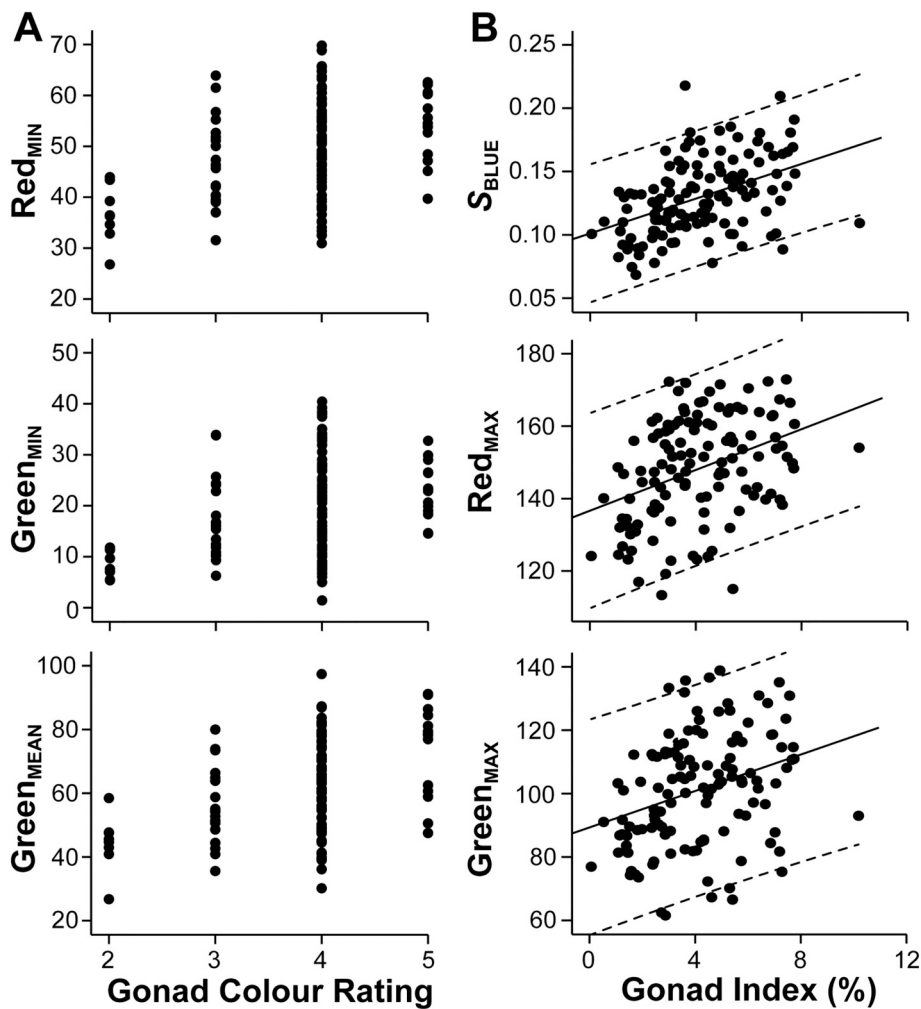


Fig. 1. The relationship between spine colour and (A) gonad colour rating and (B) gonad index of the sea urchin *Tripneustes gratilla*. Spine colour is presented as RGB variables measured from photographs (see Methods for details). Gonad colour rating refers to a subjective colour rating from 1 to 5, with 1 representing white and 5 representing intense orange (see Methods for details). (B) Solid lines are the outcome of Pearson's correlation analyses. Dashed lines are bias-corrected and accelerated (BCa) bootstrap 95% confidence intervals. $n = 126$.

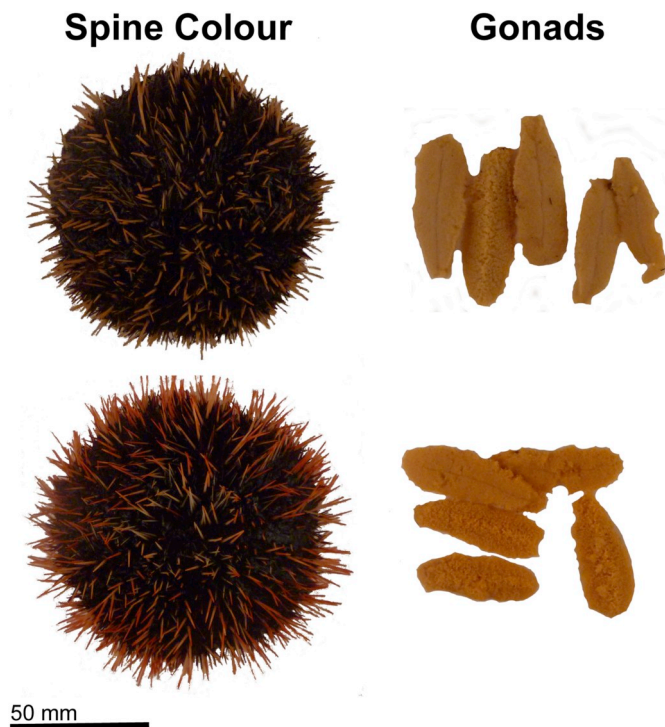


Fig. 2. Representative external and gonad morphologies of *Tripneustes gratilla*. Individuals with dark spine colours (top) tended to have low grade gonads with poor colouration, while individuals with bright spine colours (bottom) tended to have high grade gonads with excellent colouration. Individuals with bright spine colours also tended to have high gonad indices (not displayed as this is difficult to represent using photographs).

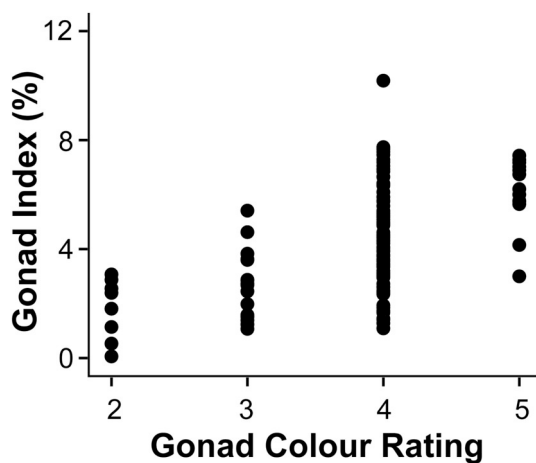


Fig. 3. There was a significant positive relationship between gonad colour rating and gonad index of the sea urchin *Tripneustes gratilla*. Gonad colour rating refers to a subjective colour rating from 1 to 5, with 1 representing white and 5 representing intense orange (see Methods for details). Spearman's rank-order correlation analysis: $r_s = 0.542$, $p < .001$, $n = 126$.

(e.g. McGraw and Toomey, 2010), fish (e.g. Folkestad et al., 2008), mammals (e.g. Stahl et al., 1998), crustaceans (e.g. Wade et al., 2005), and plants (e.g. Phan and Hsu, 1973). Despite this well documented relationship, this study is the first to report a correlation between internal and external colouration for any sea urchin. The relationship we found may have been because the colour of the epidermis of *Tripneustes* spines is conspicuous against the white skeletal tissue beneath. Variation in spine colour is likely to be more difficult to detect in sea urchins whose skeletal tissues are strongly coloured. It would be interesting to see whether RGB or other techniques for assessing spine colour can also

be used to predict the amount or type of colour pigments in spines and other body structures, similar to the way in which astaxanthin and fat content in Atlantic salmon, *Salmo salar* can be predicted from colour measured using visible and near infrared spectroscopy (Folkestad et al., 2008).

A somewhat unexpected result was that *T. gratilla* with brighter coloured spines tended to have larger gonads. Similarly, the largest gonads also had the most intense colour. For sea urchins, relatively large gonads are often pale, thought to be due to a relative decrease in the concentration of carotenoids as gonads rapidly increase in size (e.g. Shpigel et al., 2005; Watts et al., 1998). However for many animals, especially those that use colour signalling in sexual selection, external colour can predict the size of gonads and/or reproductive output, for example in birds (Doucet et al., 2005; Doutrelant et al., 2008), insects (Bots et al., 2009; Nokelainen et al., 2012), anurans (Vásquez and Pfennig, 2007, but see Hettyey et al., 2009), reptiles (Vercken et al., 2007; Weiss et al., 2009) and fish (Berglund et al., 1997; Locatello et al., 2006). Likewise, higher levels of colour pigments are associated with greater reproductive output (e.g. Helfenstein et al., 2010; Ogilvy et al., 2012). A possible explanation for a relationship between external colour and gonad size is that the acquisition of colour pigments may depend on the condition of an organism ('index hypothesis', Weaver et al., 2017). *T. gratilla* that were capable of growing large gonads also may have had sufficient energy to acquire and store greater amounts of colour pigments like carotenoids and naphthoquinones. An alternate explanation is that the most colourful individuals had the largest gonads because higher levels of colour pigments provided health benefits (e.g. increased immunity, antioxidant functions, Olson and Owens, 1998). If so, we might expect that making it easier for sea urchins to acquire colour pigments would improve gonad indices. However, for all sea urchins tested thus far, providing additional carotenoids in food does not enhance gonad size (Hávardsson et al., 1999; Kennedy et al., 2007; McBride et al., 2004; Peng et al., 2012; Plank et al., 2002; Robinson et al., 2002; Shpigel et al., 2006; Suckling et al., 2011).

It is likely the patterns in colour identified by our models were reflective of variation in the types or amounts of colour pigments in epithelial and gonadal tissues. Some of the naphthoquinones found in *T. gratilla* spines, spinochrome A and 2,5,7,8-tetrahydroxy-1,4-naphthoquinone (Anderson et al., 1969) and spinochrome D – iso 3, spinochrome E, and echinochrome A (Brasseur et al., 2017), appear orange or red (Anderson et al., 1969). The gonads of *T. gratilla* contain at least 14 types of carotenoids including β -echinenone, β -carotene, (6R)- α -carotene, and (6'R)- α -echinenone (Matsuno and Tsushima, 2001; Shina et al., 1978), some of which produce orange and red colours, although the presence of these carotenoids in spines has not been assessed. Greater understanding of the role of colour pigments in spine colour and its relationship with gonad size and colour will require simultaneous measurements of carotenoids and naphthoquinones in spines and gonads.

5. Conclusions

This study is an important first step in developing a non-lethal technique for determining the harvest readiness of sea urchins. The capacity of our models to predict gonad characteristics from RGB spine colour assessments will need to be improved prior to commercial application. Ideally RGB or similar colour assessments would be automated, perhaps using machine learning and machine vision approaches integrated into harvesting systems. Automated systems that utilise colour assessments are already in development for finfish culture and harvesting (Saberioon et al., 2017). It is unknown, however, whether the links between spine colour and gonad index and colour rating are restricted to *Tripneustes*, a genus known for its highly variable spine colour. Further research will be required to assess the viability of spine colour assessments as tools to determine harvest readiness for other commercially important sea urchins that also have variable spine

colour, such as *Paracentrotus lividus*. It may also be interesting to examine whether the colour of tests or tube feet are related to gonad size or colour, as the colour of these structures can be highly variable in *T. gratilla* (Toha et al., 2015).

Research data statement

Due to the potential for commercial application, requests for raw data will be considered on a case-by-case basis.

Acknowledgments

This study was funded by Rural Industries Research and Development Corporation (RIRDC), Australia, trading as AgriFutures Australia (New Animal Products grants: Project No PRJ-006543 and PRJ-010284). The funding body had no role in the design of the study, in the collection, analysis and interpretation of data, in the writing of the article, or in the decision to submit the article for publication. We thank two anonymous reviewers whose comments improved the manuscript.

References

- Addis, P., Moccia, D., Secci, M., 2015. Effect of two different habitats on spine and gonad colour in the purple sea urchin *Paracentrotus lividus*. *Mar. Ecol.* 36, 178–184.
- Aksnes, A., Gjerde, B., Roald, S.O., 1986. Biological, chemical and organoleptic changes during maturation of farmed Atlantic salmon, *Salmo salar*. *Aquaculture* 53, 7–20.
- Anderson, M.J., 2001. A new method for non-parametric multivariate analysis of variance. *Austral. Ecol.* 26, 32–46.
- Anderson, H.A., Mathieson, J.W., Thomson, R.H., 1969. Distribution of spinochrome pigments in echinoids. *Comp. Biochem. Physiol.* 28, 333–345.
- Anderson, M.J., Gorley, R.N., Clarke, K.R., 2008. PERMANOVA+ for Primer: Guide to Software and Statistical Methods. PRIMER-E Ltd, Plymouth.
- Andrew, N.L., Agatsuma, Y., Ballesteros, E., Bazhin, A.G., Creaser, E.P., Barnes, D.K.A., Botsford, L.W., Bradbury, A., Campbell, A., Dixon, J.D., Einarsson, S., Gerring, P.K., Hebert, K., Hunter, M., Hur, S.B., Johnson, C.R., Juinio-Menez, M.A., Kalvass, P., Miller, R.J., Moreno, C.A., Palleiro, J.S., Rivas, D., Robinson, S.M.L., Schroeter, S.C., Steneck, R.S., Vadas, R.L., Woodby, D.A., Xiaoqi, Z., 2002. Status and management of world sea urchin fisheries. *Oceanogr. Mar. Biol.* 40, 343–425.
- Azad, A.K., Pearce, C.M., McKinley, R.S., 2011. Effects of diet and temperature on ingestion, absorption, assimilation, gonad yield, and gonad quality of the purple sea urchin (*Strongylocentrotus purpuratus*). *Aquaculture* 317, 187–196.
- Berglund, A., Rosenqvist, G., Bernet, P., 1997. Ornamentation predicts reproductive success in female pipefish. *Behav. Ecol. Sociobiol.* 40, 145–150.
- Bots, J., Van Dongen, S., Adriaens, T., Dumont, H.J., Stoks, R., Van Gossum, H., 2009. Female morphs of a colour polymorphic damselfly differ in developmental instability and fecundity. *Anim. Biol.* 59, 41–54.
- Brasseur, L., Hennebert, E., Fievez, L., Caulier, G., Bureau, F., Tafforeau, L., Flammang, P., Gerbaux, P., Eckhaut, I., 2017. The roles of spinochromes in four shallow water tropical sea urchins and their potential as bioactive pharmacological agents. *Marine Drugs* 15.
- Brown, N.P., Eddy, S.D., 2015. *Echinoderm Aquaculture*. Wiley-Blackwell, Hoboken, New Jersey, USA.
- Brown, A., Patlan, D., 1974. Color changes in the ovaries of penaeid shrimp as a determinant of their maturity. *Mar. Fish. Rev.* 36, 23–26.
- Cott, H.B., 1940. *Adaptive Coloration in Animals*. Methuen & Co Ltd, Strand, London.
- Cyrus, M.D., Bolton, J.J., Macey, B.M., 2015. The role of the green seaweed *Ulva* as a dietary supplement for full life-cycle grow-out of *Tripneustes gratilla*. *Aquaculture* 446, 187–197.
- Dean, T.A., Schroeter, S.C., Dixon, J.D., 1984. Effects of grazing by two species of sea urchins (*Strongylocentrotus franciscanus* and *Lytechinus anamesus*) on recruitment and survival of two species of kelp (*Macrocystis pyrifera* and *Pterygophora californica*). *Mar. Biol.* 78, 301–313.
- Dong, S., Liang, M., Gao, Q., Wang, F., Dong, Y., Tian, X., 2010. Intra-specific effects of sea cucumber (*Apostichopus japonicus*) with reference to stocking density and body size. *Aquac. Res.* 41, 1170–1178.
- Doucet, S.M., Mennill, D.J., Montgomerie, R., Boag, P.T., Ratcliffe, L.M., 2005. Achromatic plumage reflectance predicts reproductive success in male black-capped chickadees. *Behav. Ecol.* 16, 218–222.
- Doutrelant, C., Grégoire, A., Grnac, N., Gomez, D., Lambrechts, M.M., Perret, P., 2008. Female coloration indicates female reproductive capacity in blue tits. *J. Evol. Biol.* 21, 226–233.
- Endler, J.A., 1980. Natural selection on colour patterns in *Poecilia reticulata*. *Evolution* 34, 76–91.
- Folkestad, A., Wold, J.P., Rørvik, K.-A., Tschudi, J., Haugholt, K.H., Kolstad, K., Mørkøre, T., 2008. Rapid and non-invasive measurements of fat and pigment concentrations in live and slaughtered Atlantic salmon (*Salmo salar* L.). *Aquaculture* 280, 129–135.
- Fox, H.M., Vevers, G., 1960. *The Nature of Animal Colours*. Macmillan, New York.
- Gillespie, A.R., Kahle, A.B., Walker, R.E., 1987. Color enhancement of highly correlated images. II. Channel ratio and “chromaticity” transformation techniques. *Remote Sens. Environ.* 22, 343–365.
- Grosjean, P., Spirlet, C., Gosselin, P., Vaitilingon, D., Jangoux, M., 1998. Land-based, closed-cycle echiniculture of *Paracentrotus lividus* (Lamarck) (Echinoidea : Echinodermata): a long-term experiment at a pilot scale. *J. Shellfish Res.* 17, 1523–1531.
- Hávardsson, B., Imsland, A.K., Christiansen, R., 1999. The effect of astaxanthin in feed and environmental temperature on carotenoid concentration in the gonads of the green sea urchin *Strongylocentrotus droebachiensis* Müller. *J. World Aquacult. Soc.* 30, 208–218.
- Heflin, L.E., Makowsky, R., Taylor, J.C., Williams, M.B., Lawrence, A.L., Watts, S.A., 2016. Production and economic optimization of dietary protein and carbohydrate in the culture of juvenile sea urchin *Lytechinus variegatus*. *Aquaculture* 463, 51–60.
- Helfenstein, F., Losdat, S., Möller, A.P., Blount, J.D., Richner, H., 2010. Sperm of colourful males are better protected against oxidative stress. *Ecol. Lett.* 13, 213–222.
- Hettty, A., Herczeg, G., Hoi, H., 2009. Testing the phenotype-linked fertility hypothesis in male Moor Frogs (*Rana arvalis*) exhibiting a conspicuous nuptial coloration. *Amphibia-Reptilia* 30, 581–586.
- James, P.J., 2006. A comparison of roe enhancement of the sea urchin *Evechinus chloroticus* in sea-based and land-based cages. *Aquaculture* 253, 290–300.
- Jobling, M., Baardvik, B.M., 1994. The influence of environmental manipulations on inter- and intra-individual variation in food acquisition and growth performance of Arctic charr, *Salvelinus alpinus*. *J. Fish Biol.* 44, 1069–1087.
- Juinio-Menez, M.A., Bangi, H.G., Malay, M.C., Pastor, D., 2008. Enhancing the recovery of depleted *Tripteneustes gratilla* stocks through grow-out culture and restocking. *Rev. Fish. Sci.* 16, 35–43.
- Kennedy, E.J., Robinson, S.M.C., Parsons, G.J., Castell, J.D., 2007. Effect of dietary minerals and pigment on somatic growth of juvenile green sea urchins, *Strongylocentrotus droebachiensis*. *J. World Aquacult. Soc.* 38, 36–48.
- Laizon-Guay, J.-S., Scheibling, R.E., 2007. Behaviour of sea urchin *Strongylocentrotus droebachiensis* grazing fronts: food-mediated aggregation and density-dependent facilitation. *Mar. Ecol. Prog. Ser.* 329, 191–204.
- Lewis, J.B., Storey, G.S., 1984. Differences in morphology and life history traits of the echinoid *Echinometra lucunter* from different habitats. *Mar. Ecol. Prog. Ser.* 15, 207–211.
- Locatello, L., Rasotto, M.B., Evans, J.P., Pilastro, A., 2006. Colourful male guppies produce faster and more viable sperm. *J. Evol. Biol.* 19, 1595–1602.
- Lozano, G.A., 1994. Carotenoids, parasites, and sexual selection. *Oikos* 70, 309–311.
- Matsuno, T., Tushima, M., 2001. Carotenoids in sea urchins. In: Lawrence, J.M. (Ed.), *Developments in Aquaculture and Fisheries Science*. Elsevier, Amsterdam, pp. 115–138.
- McArdle, B., Anderson, M., 2001. Fitting multivariate models to community data: a comment on distance-based redundancy analysis. *Ecology* 82, 290–297.
- McBride, S.C., Price, R.J., Tom, P.D., Lawrence, J.M., Lawrence, A.L., 2004. Comparison of gonad quality factors: color, hardness and resilience, of *Strongylocentrotus franciscanus* between sea urchins fed prepared feed or algal diets and sea urchins harvested from the Northern California fishery. *Aquaculture* 233, 405–422.
- McGraw, Kevin J., Toomey, Matthew B., 2010. Carotenoid accumulation in the tissues of zebra finches: predictors of integumentary pigmentation and implications for carotenoid allocation strategies. *Physiol. Biochem. Zool.* 83, 97–109.
- Mos, B., Cowden, K.L., Nielsen, S.J., Dworjanyn, S.A., 2011. Do cues matter? Highly inductive settlement cues don't ensure high post-settlement survival in sea urchin aquaculture. *PLoS One* 6, e28054.
- Mos, B., Byrne, M., Cowden, K.L., Dworjanyn, S.A., 2015. Biogenic acidification drives density-dependent growth of a calcifying invertebrate in culture. *Mar. Biol.* 162, 1541–1558.
- Mos, B., Byrne, M., Dworjanyn, S.A., 2016. Biogenic acidification reduces sea urchin gonad growth and increases susceptibility of aquaculture to ocean acidification. *Mar. Environ. Res.* 113, 39–48.
- Nokelainen, O., Hegna, R.H., Reudler, J.H., Lindstedt, C., Mappes, J., 2012. Trade-off between warning signal efficacy and mating success in the wood tiger moth. *Proc. R. Soc. B Biol. Sci.* 279, 257–265.
- Ogilvy, V., Preziosi, R.F., Fidgett, A.L., 2012. A brighter future for frogs? The influence of carotenoids on the health, development and reproductive success of the red-eye tree frog. *Anim. Conserv.* 15, 480–488.
- Olson, V.A., Owens, I.P.F., 1998. Costly sexual signals: are carotenoids rare, risky or required? *Trends Ecol. Evol.* 13, 510–514.
- Pearce, C.M., Daggett, T.L., Robinson, S.M.C., 2003. Effects of starch type, macroalgal meal source, and beta-carotene on gonad yield and quality of the green sea urchin, *Strongylocentrotus droebachiensis* (Müller), fed prepared diets. *J. Shellfish Res.* 22, 505–519.
- Peng, J., Yuan, J.-P., Wang, J.-H., 2012. Effect of diets supplemented with different sources of astaxanthin on the gonad of the sea urchin *Anthodidaris crassispina*. *Nutrients* 4, 922.
- Phan, C.T., Hsu, H., 1973. Physical and chemical changes occurring in the carrot root during growth. *Can. J. Plant Sci.* 53, 629–634.
- Plank, L.R., Lawrence, J.M., Lawrence, A.L., Olvera, R.M., 2002. The effect of dietary carotenoids on gonad production and carotenoid profiles in the sea urchin *Lytechinus variegatus*. *J. World Aquacult. Soc.* 33, 127–137.
- Qi, S., Zhang, W., Jing, C., Wang, H., Zhao, S., Zhou, M., Chang, Y., 2016. Long-term effects of stocking density on survival, growth performance and marketable production of the sea urchin *Strongylocentrotus intermedius*. *Aquac. Int.* 24, 1323–1339.
- Robinson, S.M.C., Castell, J.D., Kennedy, E.J., 2002. Developing suitable colour in the gonads of cultured green sea urchins (*Strongylocentrotus droebachiensis*). *Aquaculture* 206, 289–303.
- Rueden, C.T., Schindelin, J., Hiner, M.C., DeZonia, B.E., Walter, A.E., Arena, E.T., Eliceiri,

- K.W., 2017. ImageJ2: imageJ for the next generation of scientific image data. *BMC Bioinformatics*. 18, 529.
- Saberioon, M., Gholizadeh, A., Cisar, P., Pautsina, A., Urban, J., 2017. Application of machine vision systems in aquaculture with emphasis on fish: state-of-the-art and key issues. *Rev. Aquac.* 9, 369–387.
- Shina, A., Gross, J., Lifshitz, A., Sklarz, B., 1978. Carotenoids of the invertebrates of the red sea (Eilat shore)- II carotenoid pigments in the gonads of the sea urchin *Triploneustes gratilla* (Echinodermata). *Comp. Biochem. Physiol. B* 61, 123–128.
- Shpigel, M., McBride, S.C., Marciano, S., Ron, S., Ben-Amotz, A., 2005. Improving gonad colour and somatic index in the European sea urchin *Paracentrotus lividus*. *Aquaculture*. 245, 101–109.
- Shpigel, M., Schlosser, S.C., Ben-Amotz, A., Lawrence, A.L., Lawrence, J.M., 2006. Effects of dietary carotenoid on the gut and the gonad of the sea urchin *Paracentrotus lividus*. *Aquaculture*. 261, 1269–1280.
- Shpigel, M., Shauli, L., Odintsov, V., Ashkenazi, N., Ben-Ezra, D., 2018. *Ulva lactuca* biofilter from a land-based integrated multi trophic aquaculture (IMTA) system as a sole food source for the tropical sea urchin *Triploneustes gratilla elatensis*. *Aquaculture*. 496, 221–231.
- Stahl, W., Heinrich, U., Jungmann, H., von Laar, J., Schietzel, M., Sies, H., Tronnier, H., 1998. Increased dermal carotenoid levels assessed by noninvasive reflection spectrophotometry correlate with serum levels in women ingesting betatene. *J. Nutr.* 128, 903–907.
- Suckling, C.C., Symonds, R.C., Kelly, M.S., Young, A.J., 2011. The effect of artificial diets on gonad colour and biomass in the edible sea urchin *Psammochinus miliaris*. *Aquaculture*. 318, 335–342.
- Toha, A.H.A., Sumitro, S.B., Widodo Hakim, L., 2015. Color diversity and distribution of sea urchin *Triploneustes gratilla* in Cenderawasih Bay ecoregion of Papua, Indonesia. *Egypt. J. Aquatic Res.* 41, 273–278.
- Tsushima, M., 2007. Carotenoids in sea urchins. In: Lawrence, J.M. (Ed.), *Edible Sea Urchins: Biology and Ecology*. Elsevier Science, Amsterdam, pp. 159–166.
- Unuma, T., 2002. Gonadal growth and its relationship to aquaculture in sea urchins. In: Yokota, Y., Matranga, V., Smolenicka, Z. (Eds.), *The Sea Urchin: From Basic Biology to Aquaculture*. A. A. Balkema Publishers, Lisse, pp. 115–127.
- Vásquez, T., Pfennig, K.S., 2007. Looking on the bright side: females prefer coloration indicative of male size and condition in the sexually dichromatic spadefoot toad, *Scaphiopus couchii*. *Behav. Ecol. Sociobiol.* 62, 127–135.
- Vercken, E., Massot, M., Sinervo, B., Clobert, J., 2007. Colour variation and alternative reproductive strategies in females of the common lizard *Lacerta vivipara*. *J. Evol. Biol.* 20, 221–232.
- Wade, N., Goulter, K.C., Wilson, K.J., Hall, M.R., Degnan, B.M., 2005. Esterified astaxanthin levels in lobster epithelia correlate with shell colour intensity: potential role in crustacean shell colour formation. *Comp. Biochem. Physiol. B: Biochem. Mol. Biol.* 141, 307–313.
- Watts, S.A., Boettger, S.A., McClintock, J.B., Lawrence, J.M., 1998. Gonad production in the sea urchin *Lytechinus variegatus* (Lamarck) fed prepared diets. *J. Shellfish Res.* 17, 1591–1595.
- Weaver, R.J., Koch, R.E., Hill, G.E., 2017. What maintains signal honesty in animal colour displays used in mate choice? *Philos. Trans. Royal Soc. B: Biol. Sci.* 372.
- Weiss, S.L., Kennedy, E.A., Bernhard, J.A., 2009. Female-specific ornamentation predicts offspring quality in the striped plateau lizard, *Sceloporus virgatus*. *Behav. Ecol.* 20, 1063–1071.
- Westbrook, C.E., Ringang, R.R., Cantero, S.M.A., Toonen, R.J., 2015. Survivorship and feeding preferences among size classes of outplanted sea urchins, *Triploneustes gratilla*, and possible use as biocontrol for invasive alien algae. *PeerJ*. 3, e1235.
- Whitaker, R., Quinlan, W., Daley, C., Parsons, J., 1997. Developing markets for feed lot sea urchins. *Bull. Aquacult. Assoc. Can.* 42–44.
- Woods, C.M.C., James, P.J., Moss, G.A., Wright, J., Siikavuopio, S., 2008. A comparison of the effect of urchin size and diet on gonad yield and quality in the sea urchin *Evechinus chloroticus* Valenciennes. *Aquac. Int.* 16, 49–68.